

GRIFOLS Diagnostic Grifols, S.A.	Reference:	REGD-0002168		
	Version:	8.0; Most-Recent; Effective; CURRENT		
	Status:	Effective	Purpose:	Uncontrolled Copy
Class:	Control del Diseño		Effective date:	17-May-2022
Title:	Service manual DG SPIN			

DG SPIN

Service Manual (Edition March 2022)

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(this Service Manual starts on page 2)

This manual uses the following symbols to facilitate the location of specific sections.



DANGER!

Any kind of harmful situation for people that can be diminished using this indication or that it is convenient the QUALIFIED TECHNICIAN knows it.



WARNING!

Any kind of injure for people for whom protection systems are provided, but it is convenient the QUALIFIED TECHNICIAN knows it.



CAUTION!

Any kind of harmful situation for the equipment or other materials. This situation can be diminished using this indication or with other protections.



NOTE:

Used for clarification and complementary or emphasized information.

RESPONSIBLE AUTHORITY

Individual or group responsible for the use and maintenance of the equipment, and for ensuring that operators are adequately trained.

OPERATOR

Person operating the equipment for its intended purpose. The operator should have received training appropriate for this purpose.

QUALIFIED TECHNICIAN

Person responsible for the installation, reparation and special maintenance of the equipment who has received specific training for this purpose.

All restrictions referred to the RESPONSIBLE AUTHORITY are valid for the OPERATOR.

This instrument is protected by international patents applicable to the entire machine or parts thereof.

The manufacturers reserve the right to take legal actions in order to protect their interests.

This manual should be used in conjunction with the Instructions for Use.

The information found in this document may be changed without prior notification.

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Changes from previous version:

Revision	Change
03/2007	N/A
09/2007	Fix error in Troubleshooting section: Message E8 must be E9.
05/2008	Update assembly and cabling drawings
12/2009	-Added a procedure to update firmware without <i>DG Terminal</i>. -Added drawing S0721V0 5/6. -Updated drawing S0803V0. -References to Safety Information and Technical Specifications are included. -Added Fuses and LEDs information. -Added Power Supply information. -Added Regulation Warning problem in Troubleshooting -Added information about 'RegCentr.txt' -Updated images. -Minor typographic errors.
11/2010	- Added E2 error message to Troubleshooting section. - Added new warning message NO ROTOR PRESENCE. - Added Centrifugation without rotor case in Troubleshooting section. - Included information about time basis of the centrifugation. - Updated drawing S0721V0.
07/2011	- Updated electronic schematics and mechanical drawings. - Added continuity measurement after corrective actions. - Added warning in chapters 2 and 6.
04/2021	- Document format updated. - Added diagram for internal AC power cabling connection. - Added Creo View Express program in Drawings section. - Updated mechanical drawings. - Removed electronic schematics. - Minor text modifications.
03/2022	- Updated section 6.1 <i>Troubleshooting</i>. Communication errors during firmware update added. Warning note for the electric continuity test added. - Updated section 6.2.1 <i>Updating the firmware (firmware version ≥ 1.3.1 and DG Terminal ≥ v1.3.0)</i>. Firmware update procedure updated for firmware version ≥ 1.3.1 and DG Terminal v1.3.0. - Updated section 8 <i>Drawings</i>. Creo View Express minimum software version added. Mechanical drawings table updated. - Minor text modifications.

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1 INSTRUMENT GENERAL OVERVIEW

This manual is oriented to the **QUALIFIED TECHNICIAN** and contains the necessary information for a proper and secure manipulation of the instrument.

Read carefully all the information of this manual before starting the manipulation of the instrument. It is important to read the information concerning maintenance, correct use and liquids management.



NOTE:

All the operations described in this manual must be carried out only by a **QUALIFIED TECHNICIAN**.

1.1 Overview

The *DG Spin* centrifuge has been especially designed to carry out centrifugation of a maximum of 24 gel cards in one sole batch. Therefore, the predefined parameters in the instrument (speed and time) are those required to work with this type of card.

1.2 Parts

The *DG Spin* main parts are the following:

- *DG Spin head*: the mobile part that holds the cards. It includes the polycarbonate disk, 24 card holders and the handler. The disk has a reflective tape to measure the centrifuge speed with an optical tachometer.
- *Motor*: the brushless DC motor including the rotor holder bushing, and the silent blocks.
- *Electromagnetic lock*: security closure that detects the aperture of the lid.
- *Electronics*: the different PCAs, power supply, power inlet, display and keypad.
- *Enclosure*: includes the case, the lid and the base.

Drawings of *DG Spin* are listed in section 8.2 *Drawings reference table*. Assembly drawings are self-explanatory and they may include notes for manufacturing purposes that should not be considered.

1.3 Electronics

Purpose and function of each PCA and related parts is described below.

1.3.1 Mains socket with filter

1.3.1.1 Fuses

There are 2 fuses in the mains socket with filter:

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- FUS1 → F2A/250V
- FUS2 → F2A/250V

1.3.1.2 Power Supply

Adapts the AC supply voltage into DC 48V. This power supply automatically adapts the input AC voltage (AC 85-264V), so it does not require any type of manual switching.

Internal AC power cabling between the Mains filter and the power supply is color coded, make sure the connection is done according to the diagram below:

- Brown: Line (1) (P)
- Light blue: Neutral (2) (N)
- Yellow-green: Ground (3) (PE)

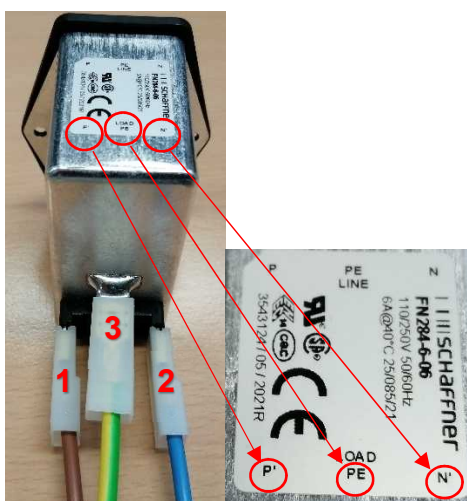


Figure 1. Mains filter connection position identification

1.3.2 PCA DC Motor Control

The PCA DC Motor Control is a generic PCA to control motor. Containing the following key components:

- Power supply: converts the incoming +5V supply to +3V3, 2V5 and 1V2 needed to power the digital circuits.
- 64-pin DIN41612 connector: connects the PCA DC Motor Control with the PCA Distribution, described below.
- Digital circuits: with the Renesas microcontroller as the main component, housing the control firmware, a non-volatile memory to store the configuration parameters and a RS232 interface for technical service tasks.
- FPGA: serves to configure the control of each motor driver confirming the type of DC motor used. Furthermore, it is used to interconnect I/Os of the connector and I/Os of the microcontroller.
- 3 motor drivers: capable to drive up to 3 brushless or 4 brushed DC motors. In the present design only one motor driver is used (driver 1), as only one brushless motor is used.

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- Contains a crystal (clock used as a time basis for centrifugation process)

1.3.2.1 LEDs

The PCA DC Motor Control has 2 LEDs:

- LED1 → When the *DG Spin* is switched on, the LED is ON and a few seconds later is OFF. This means that the FPGA is loaded correctly with the appropriate data.
- LED2 → Indicates that the voltage supplied to the motor is $\geq 12V$. Notice that the motor voltage is 48V, so it is necessary to measure the voltage to confirm that the 48V are ok.

1.3.2.2 Fuses

The PCA DC Motor Control has 1 fuse:

- FUS1 → Fuse MSF F800mA / 250V

1.3.3 PCA Distribution

The PCA Distribution contains the following circuits:

- Connectors: Connections for all external parts, including:
 - The display
 - The electromagnetic lock
 - The brushless motor
 - The vibration sensor
 - The PCA DC Motor control
- Lid lock: Serves for firing the lock and detection of the lock state (open/close).
- Power supply: Converts the incoming +48V supply to +5V.
- RS232-RS485: Communication interface that converts the incoming RS232 to the RS485 protocol.

1.3.3.1 Fuses

The PCA Distribution has 1 fuse:

- FUS1 → Fuse MSF F1A / 250V

1.3.4 PCA Unbalance Detector

Used to detect excessive vibration of the *DG Spin* by means of an accelerometer.

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1.3.5 PCA Display

The *DG Spin* uses a custom display to show the timer and velocity information and the incidences that could occur during the operation. The PCA Display contains a LCD with a backlight, a LCD Driver, a real time clock (clock used as a time basis for centrifugation process) and a buzzer. A two buttons keypad is connected to the PCA Display.

1.3.6 Cabling

The *DG Spin* has the following cables connections:

Number	Name	Connect
1	Cable JST XA 1496-9-068	PCA Distribution to PCA Display
2	Lock cable 1800-5-068	PCA Distribution to lid lock
3	Motor cable DG-320	PCA Distribution to motor
4	Unbalance cable DG-320	PCA Distribution to PCA Unbalance Detector
5	AC supply cable DG-320	Inlet to Power Supply
6	DC supply cable DG-320	Power Supply to PCA Distribution
7	Ground cable DG-320	Power Supply to ground

Table 1. Cabling list of centrifuge *DG Spin*

1.4 Technical specifications

See chapter 3 “*Technical Specifications*” in the “*DG Spin – Instructions for Use*”.

2 SAFETY INFORMATION

Safety Information related to the *DG Spin* is included in chapter 2 “*Safety Information*” in the “*DG Spin – Instructions for Use*”.



WARNING!

To completely disconnect the apparatus from mains remove the mains plug from the socket outlet or the appliance coupler.

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3 INSTALLATION OF DG SPIN

3.1 Unpacking

See section 5.3 *“Unpacking the equipment”* in the *“DG Spin – Instructions for Use”*.

3.2 Set-up

See chapter 5 *“Installation”* in the *“DG Spin – Instructions for Use”*.

3.3 IQ

See section 5.1 *“Installation Requirements”* in the *“DG Spin – Instructions for Use”*.

3.4 OQ

See chapter 8 *“Operational Qualification protocol (OQ)”* in the *“DG Spin – Instructions for Use”*.

3.5 Uninstalling – Shipping

See chapter 9 *“Transport and storage”* in the *“DG Spin – Instructions for Use”*.

3.6 Decontamination

See section 7.1.2 *“Decontamination of the equipment”* in the *“DG Spin – Instructions for Use”*.

3.7 Storage

See chapter 9 *“Transport and storage”* in the *“DG Spin – Instructions for Use”*.

3.8 Disposal

With the equipment properly decontaminated, disassemble all the parts.

Dispose of the different parts depending on the kind of material:

- Base:
 - Remove the bupoms: rubber.
 - Remove cable guides: plastic PA.
 - Base: stainless steel.
- Screws: stainless steel.

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- PCA DC Motors: electronic waste.
- PCA Distribution: electronic waste.
- AC Filter: electronic waste.
- Power supply: electronic waste.
- Cabling: electronic waste.
- PCA Display:
 - Remove the battery (yellow box): batteries (ion lithium).
 - Display holders: stainless steel.
 - PCA Display: electronic waste.
- Motor:
 - Remove the motor cover: plastic PS V0.
 - Remove the PCA Unbalance Detector: electronic waste.
 - Remove the shock absorbers: rubber.
 - Motor: electronic waste.
- Keypad: electronic waste.
- Lid:
 - Remove the joint: silicone.
 - Remove the metallic parts: stainless steel.
 - Lid: plastic PC V0.
- Case:
 - Remove the spring: stainless steel.
 - Remove the axis: stainless steel.
 - Case: plastic ABS+PC V0.
- Display window: plastic PC V0.
- Header:
 - Remove handle: plastic PETP.
 - Remove holders: plastic PC+GF.
 - Disc: plastic PC.

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4 DG Terminal software

The Technical Service software *DG Terminal* allows you to access to low-level functions of the *DG Spin*, in such a way that it is possible to check the individual operation of different parts of the machine, install new firmware versions, set the date and time of the real time clock or retrieve the log information of the machine, among others.

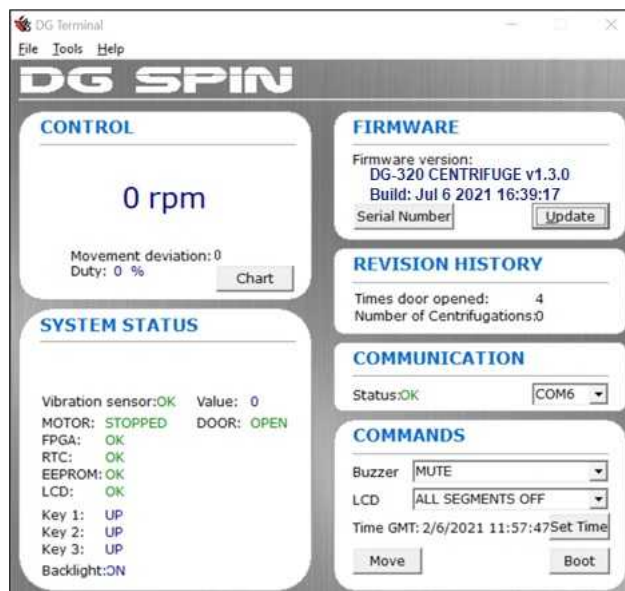


Figure 2. DG Terminal software main screen

4.1 CONTROL window

Shows all useful information about the motor control system:

- Speed in revolutions per minute (rpm).
- Deviation of the movement.
- The duty (% power) applied to the motor.

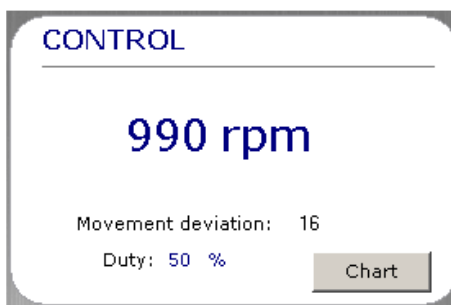


Figure 3. CONTROL window

When the Chart button is selected, a graph is shown with the applied duty and the measured speed versus time in seconds.

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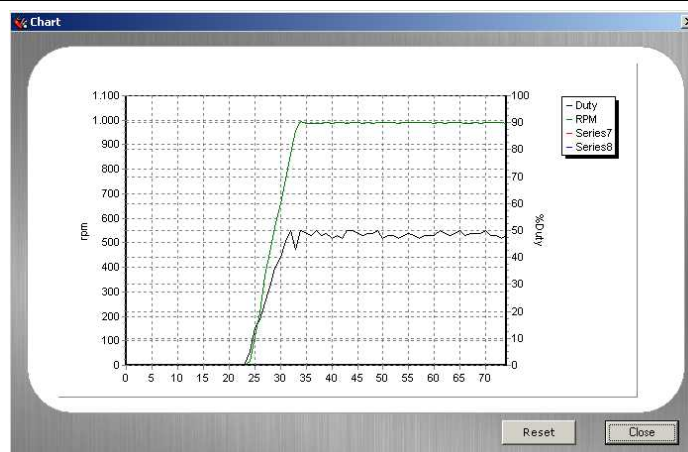


Figure 4. Chart screen

When the *DG Spin* is switched on and the *DG Terminal* is executed, the file 'RegCentr.txt' is created in the same folder where the *DG Terminal* software is located. The structure of each line of this file is:

Duty [%]; Speed [rpm]

'RegCentr.txt' is updated every second and there is not any limit for the size of the file. This file is reset pressing the *Reset* button located in the Chart window (Figure 3) or each time the *DG Terminal* is executed.

4.2 SYSTEM STATUS window

Shows the condition of different parts of the centrifuge.

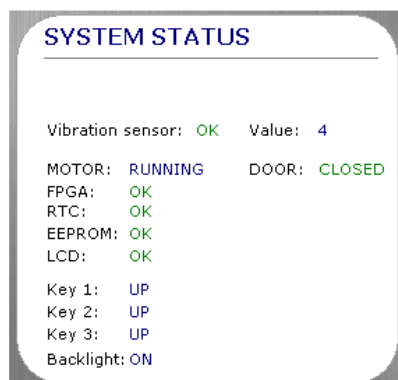


Figure 5. SYSTEM STATUS window

In case of mal functioning of the device, the origin of the incidence will be reported in this window.

4.3 FIRMWARE window

Shows the actual firmware version installed into the flash memory of the microcontroller.

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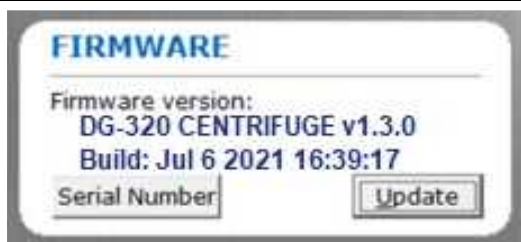


Figure 6. FIRMWARE window

4.3.1 Update firmware command

Use the Update Firmware command from the File menu or the Update button from the FIRMWARE window to install a new firmware version into the flash memory of the microcontroller.



Figure 7. Update Firmware command in File menu

4.3.2 Serial Number command

Use the serial number command to enter and store the serial number of the device into the non volatile memory of the device.

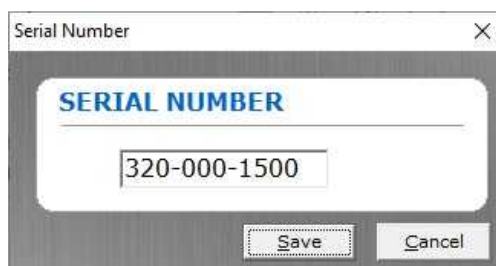


Figure 8. SERIAL NUMBER window

When no serial number is saved yet into the non volatile memory the number shown on the screen will be: 320-XXX-XXXX.

The serial number is entered at the beginning of the manufacturing process and it needs to be reentered when replacing the PCA DC Motor Control.

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4.4 REVISION HISTORY window

Shows the number of centrifugations and the times the door has been opened.



Figure 9. REVISION HISTORY window

4.5 COMMUNICATION window

Shows the serial communication status of the device.



Figure 10. COMMUNICATION window

Select the appropriate COM port if the status is not ok.

4.6 COMMANDS window

Within the commands window several subsystems of the device can be adjusted or checked.

4.6.1 Buzzer command

For testing the functionality of the buzzer, different buzzer tones can be selected.

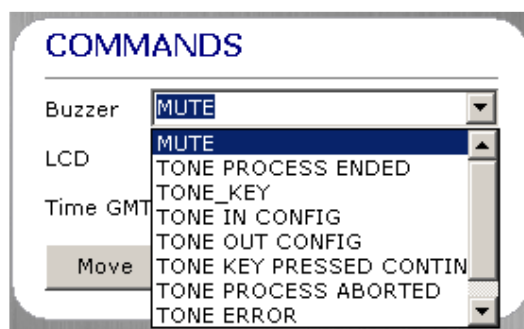


Figure 11. Buzzer options in COMMANDS window

4.6.2 LCD command

For testing the functionality of the display, all segments of the display can be switched on or off.

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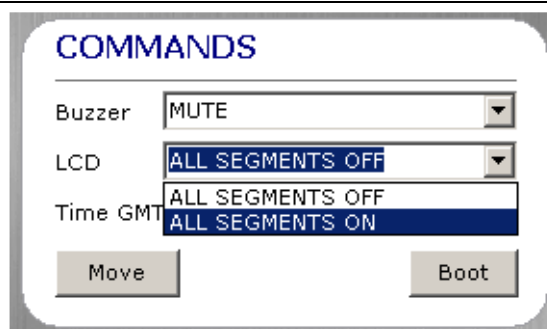


Figure 12. LCD options in COMMANDS window

4.6.3 Set Time command

For adjusting the date and time of the real time clock. The Time is stored in the device in GMT format (shown in the main window) and the *DG Terminal* converts from local time to update it.



Figure 13. Set Time window

4.6.4 Move motor command

For testing the performance of the motor, specifying the velocity and acceleration time

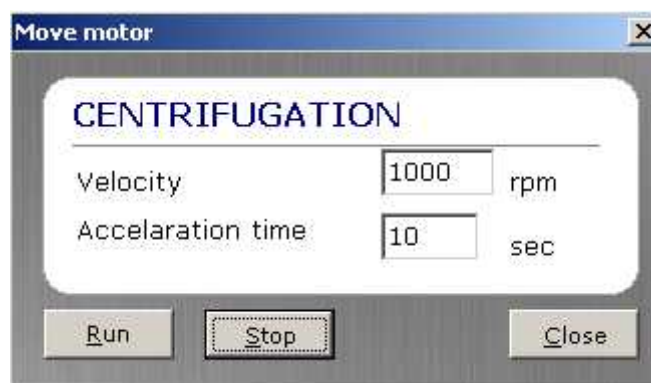


Figure 14. Move motor window

4.6.5 Boot command

Resets the device.

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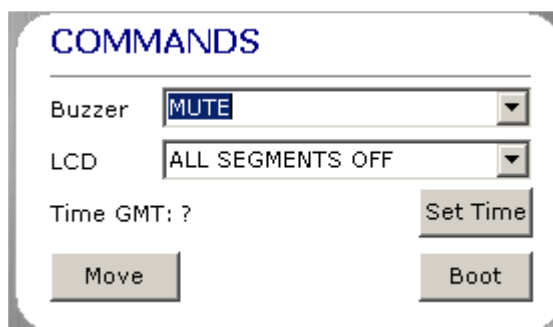


Figure 15. Boot button in COMMANDS window

4.7 Tools menu options

All events during operation of the device are stored into the non volatile memory. To retrieve the stored event information, select within the tool menu, the Log viewer option.



Figure 16. Log viewer command in Tools menu

If not already performed, data should be loaded from the non volatile memory.

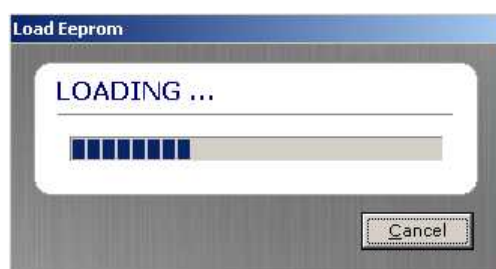


Figure 17. Loading EEPROM window

A screen with all log information will be shown, with an indication of the date and time for each event.

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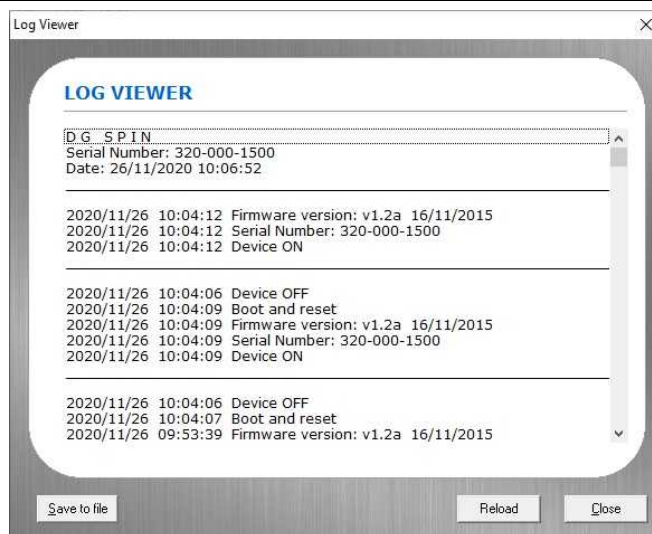


Figure 18. Log Viewer screen

To save the log information into a file, use the Save to file button.

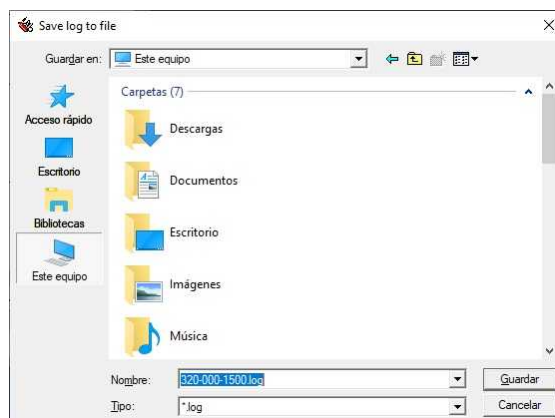


Figure 19. Save to file screen

5 PREVENTIVE MAINTENANCE

The maintenance of the *DG Spin* centrifuge only includes cleaning and decontamination: See chapter 7 “Maintenance” in the “*DG Spin – Instructions for Use*”.

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6 CORRECTIVE MAINTENANCE

6.1 Troubleshooting

INDICATION	DESCRIPTION	POSSIBLE CAUSES	SOLUTION
-	The LCD does not light up, the buzzer is off and the keypad does not work.	Power supply is defective.	Replace the power supply.
		The fuses have blown.	Replace the fuses.
		PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.
		PCA Distribution does not work properly.	Replace the PCA Distribution.
-	The LCD does not light up, the buzzer is making noise.	Display cable is damaged or badly connected.	Replace or connect the display cable.
		PCA Display does not work properly.	Replace the PCA Display.
		The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor control.
-	The Keypad does not work properly.	The keypad is badly connected or damaged.	Replace or connect the keypad.
-	A firmware update cannot be performed because communication errors have been detected.	Communication error with DG Terminal.	Check that the communications cable is well attached to both the PC and DG Spin instrument. Close DG Terminal, open it again and try to update the firmware. If the problem persists, replace the cable, RS232-USB converter (if used) or try with another PC.
	Cover opening error.	The cover is blocked.	Check if any obstacle is blocking the cover.
		The lock is damaged.	Replace the lock.
		The lock cable is damaged or badly connected.	Replace or connect the lock cable.
		The PCA Distribution does not work properly.	Replace the PCA Distribution.
		The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.
E1 	The Real Time Clock does not work properly.	The display cable is damaged or badly connected.	Replace or connect the display cable.
		The PCA Display does not work properly.	Replace the PCA Display.
		The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.

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E2 	I2C communication fails with the LCD	The PCA DC Motor Control does not work properly	Replace PCA DC Motor control.
		Display cable is damaged or badly connected.	Replace or connect the display cable.
		PCA Display does not work properly.	Replace PCA Display.
E3 	The non volatile memory does not work properly.	The PCA DC Motor Control does not work properly	Replace the PCA DC Motor Control.
E9 	The FPGA could not be programmed.	The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.
 	Motor current limit.	The motor is blocked.	Check if any obstacle is blocking the motor.
		The motor cable is damaged or badly connected.	Replace or connect the motor cable.
		The motor is damaged.	Replace the motor.
		The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.
  	The out-of balance sensor does not work properly.	The unbalance detector cable is damaged or badly connected.	Replace or connect the unbalance detector cable.
		The PCA Unbal-Det does not work properly.	Replace the PCA Unbalance Detector.
		The PCA DC Motor Control does not work properly.	Replace the PCA DC Motor Control.
 	Regulation warning.	Regulation speed of the motor is beyond its margins (± 50 rpm)	Replace the motor.
			Replace the PCA DC Motor Control.

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 	Centrifugation without rotor.	DG Spin has detected that the centrifugation is performed without rotor.	Replace the rotor.

Table 2. DG Spin troubleshooting

After a corrective action, perform an electric continuity test.



DANGER!

The aim of this test is to determine that there is continuity between accessible dead-metal parts of the product and the grounding pin or blade of the attachment plug. If all accessible dead metal is connected, only a single test need be performed: a visual or audible device (ohmmeter, buzzer, etc) may be used to indicate grounding continuity.

Check the electric continuity between the earth terminal of the IEC inlet and the metallic rear plate.



Figure 20. Earth terminal of the IEC inlet (left arrow) and metallic rear plate (right arrow) in DG Spin

Finally execute the OQ procedure to assure that the centrifuge works properly.

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6.2 Procedures and Records

6.2.1 Updating the firmware (firmware version $\geq$ 1.3.1 and DG Terminal $\geq$ v1.3.0)



NOTE:

Do not operate with the instrument and do not interrupt the process while a firmware download operation is being performed.



NOTE:

DG Terminal version 1.3.0 is required for a subsequent DG Spin instrument upgrade to Firmware version $\geq$ 1.3.1.

The *DG Spin* incorporates a firmware which is stored in the Renesas microprocessor's flash memory. To install a version of firmware you must have the following:

- DG Spin instrument
 - The technical service program *DG Terminal*.
 - A serial cable connected to a PC port.
 - A file with the version of firmware.
 - A file with the kernel file.
1. Connect a power cord to the power connector located at the back of the DG Spin instrument. Plug the power cord into the mains and turn on the instrument.
 2. Connect the communications cable to the instrument and to the PC where DG Terminal and the firmware and kernel files are available.
 3. Open DG Terminal and in the dropdown menu of the combo box control of the Communication area, select the port which is connected to the DG Spin instrument. A new window will appear, displaying the caption Checking communications.



Figure 21. Check communications window

4. Wait for a few seconds until the previous window is automatically closed. Communication status will be displayed as OK in the DG Terminal main window.



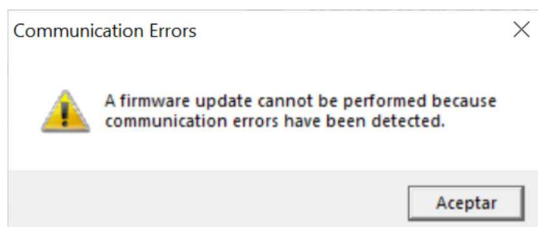
Figure 22. Communication port status

5. In the Firmware area, click on Update button.
6. If the Communications Check has determined that communications are robust enough, the Update Firmware window will be displayed.

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NOTE:

If there are communication errors, the following window will appear.
If this is the case check *Troubleshooting* section on how to proceed.



- In the Update Firmware window, click Open button, a file explorer window will appear, browse and select the kernel file and the new firmware version to be installed. Once both files have been selected, the Update button will be enabled. Click the Update button to start the installation of the new firmware.

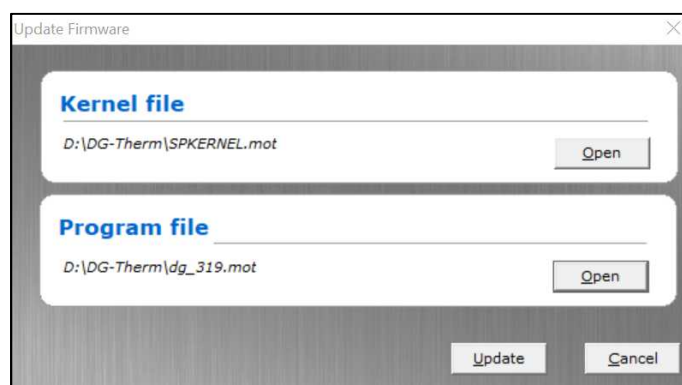


Figure 23. Update Firmware window

- The kernel program will be installed into the microcontroller.



Figure 24. SENDING KERNEL window

- After the kernel installation, it will be activated.



Figure 25. ACTIVATING KERNEL window

- The new firmware will be installed into the microcontroller flash memory.

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Figure 26. SENDING NEW FIRMWARE window

11.The device will be reset.

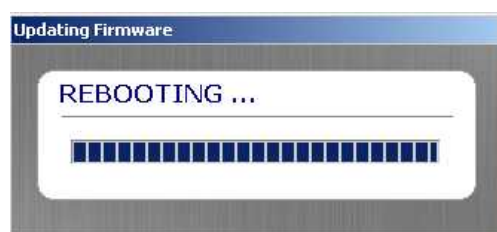


Figure 27. REBOOTING window

12.If the update process has been performed properly a confirmation will be shown on the screen. Click OK button to finish the process.

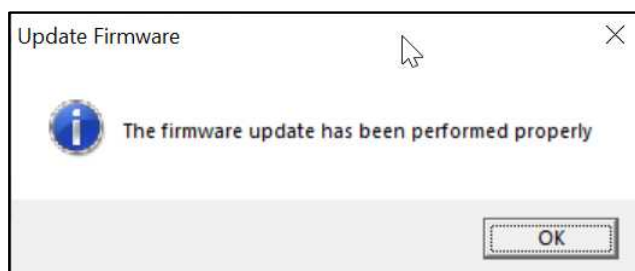


Figure 28. Update firmware process finished properly message

13.Firmware version can be checked in the Firmware area in DG Terminal.



Figure 29. Firmware version on DG Terminal

14.Switch off the instrument and disconnect the communication cable.

15.Switch on the instrument, the firmware version is displayed briefly on its display screen when the instrument is turned on.

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Figure 30. Firmware version on display

6.2.2 Updating the firmware (without DG Terminal software)

In case the firmware update fails with the *DG Terminal* software the Flash Development Toolkit (FDT) has to be used. In this case you must have the following:

- The Flash Development Toolkit 3.4 (v 3.04).
- A serial cable.
- A file with the version of *DG Spin* firmware.

6.2.2.1 Installation of the FDT

To install this software, please follow the steps described below:

- Run the Setup file fdt3_04.exe.
- Press *Next >* at the welcome screen to begin installation.
- Select the desired language and press *Next >*.
- Accept the licence terms by pressing *Yes*.
- At the following screen, select *Application* and *Kernels - Protocol B & C* and press *Next >*.



Figure 31. Select Components installation screen of FDT

NOTE: For computers with Microsoft® Windows® versions others than XP®, if the Microsoft System Files box is enabled, it must be selected.

- Press *Next >* at the additional application information screen.
- At the following screen, select *H8S/2300*, leave all other check boxes unselected and press *Next >*.

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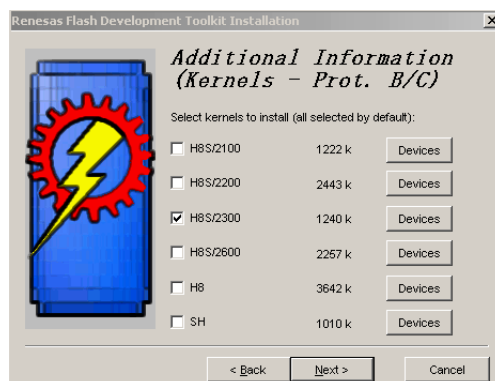


Figure 32. Additional Information (Kernels – Prot. B/C) installation screen of FDT

- Choose a destination directory and press *Next >*.
- Press *Next >* at the Select Start Menu Group screen.
- Press *Install >* at the Ready to Install screen.
- Press *Finish* when all the files are installed. The installation ends at this point.

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6.2.2.2 Installing a new firmware version on the microprocessor

To install a new firmware version on the microprocessor, connect a serial cable between the *DG Spin* and a free port on the computer and complete the following steps:

- Shortcircuit the pads of the jumper JP3 on the PCA DC Motor Control.

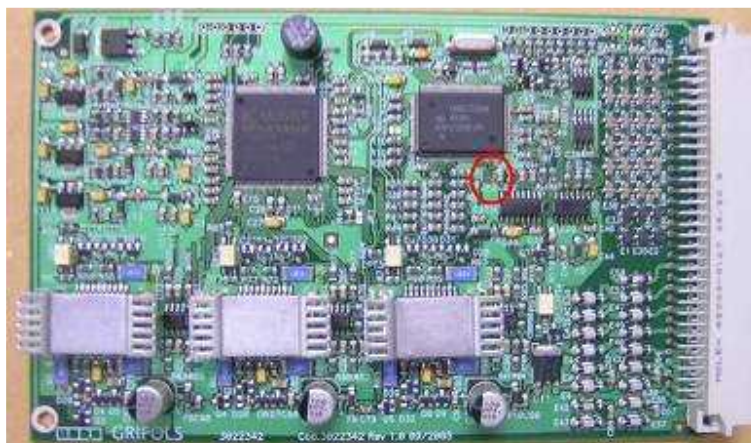


Figure 33. PCA DC Motor Control

- Connect the *DG Spin* to the mains supply.
- Run the microprocessor's FLASH programming software. The first time you open it, select the *Create a new Workspace* option and press *OK*. The Workspace that you create following the next steps will be available in the future, so you will be able to use it by double-clicking on its name the next time you open the FLASH programming software.
- Enter a name for the new workspace (e.g. DG SPIN) and click *OK*

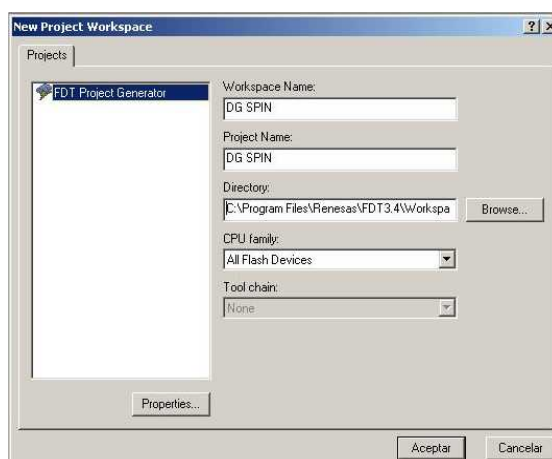


Figure 34. Workspace definition installation screen of FDT

- Choose the H8S/2328F device (Protocol B, Compiler Renesas 4.0A) or H8S/2329BF device (Protocol B, Compiler Renesas 4.0A) depending which one is installed in the PCA DC Motor Control and click *Next>*:

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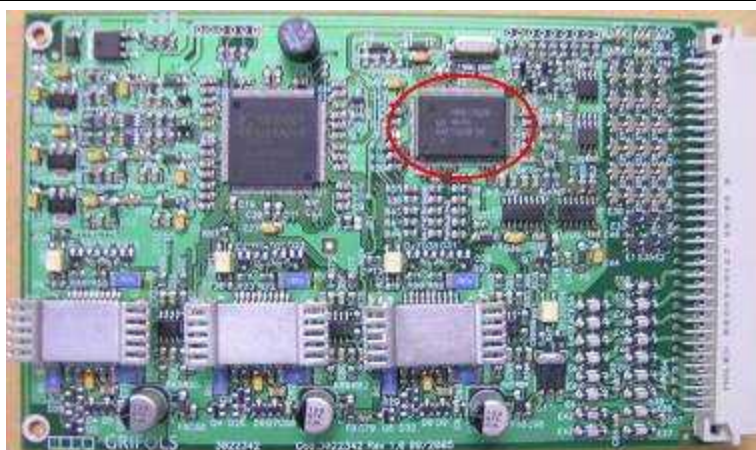


Figure 35. Location of the microcontroller in the PCA DC Motor Control

- At the following screen, select the COM port you want to employ to communicate and click *Next >*

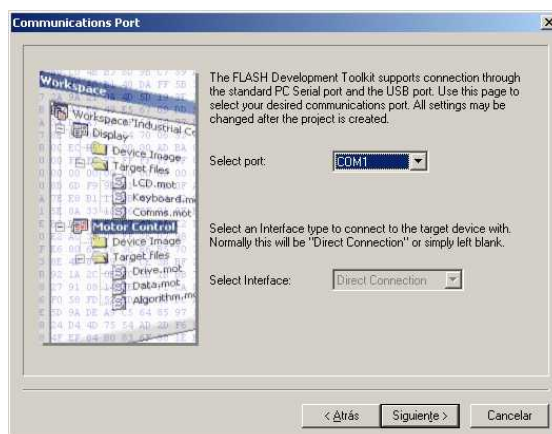


Figure 36. Communications Port installation screen of FDT

- Enter **18.432** (with the dot) for the CPU crystal frequency and click *Next >*

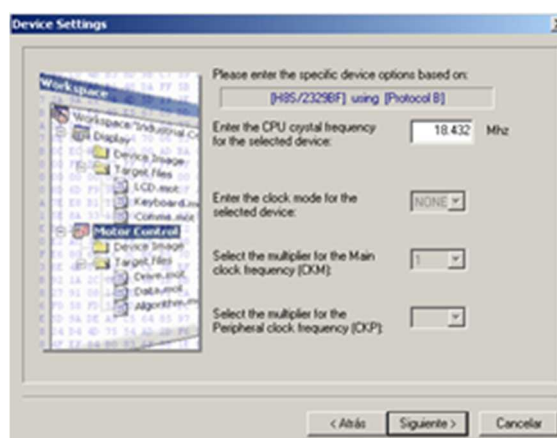


Figure 37. Device Settings installation screen of FDT

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- At the following screen, select the BOOT Mode, use default speed and click *Next* >

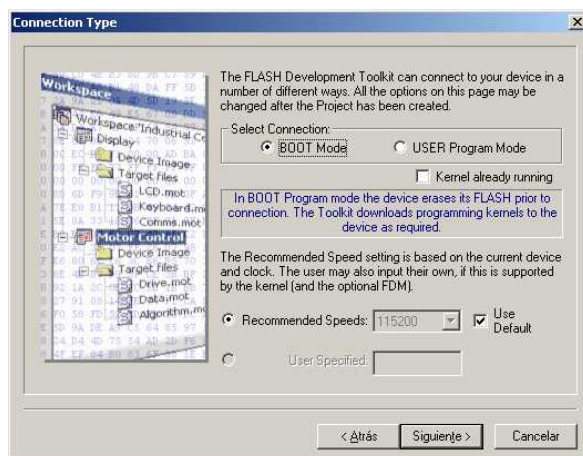


Figure 38. Connection Type installation screen of FDT

- At the following screen, select the *Interactive* level of device protection and the *Standard* level of messaging and click *Finish*.

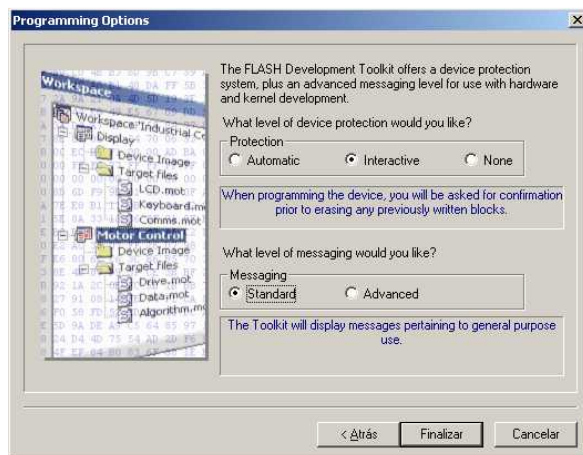


Figure 39. Programming Options installation screen of FDT

- You have successfully created a new Workspace. Now you have to add the file with the *DG Spin* firmware to the Workspace. Please note that each time you must update the current release, you'll have to erase the existing file and add the new one.
- Press the *right mouse button* on the workspace name and select the *Add Files* option at the following screen:

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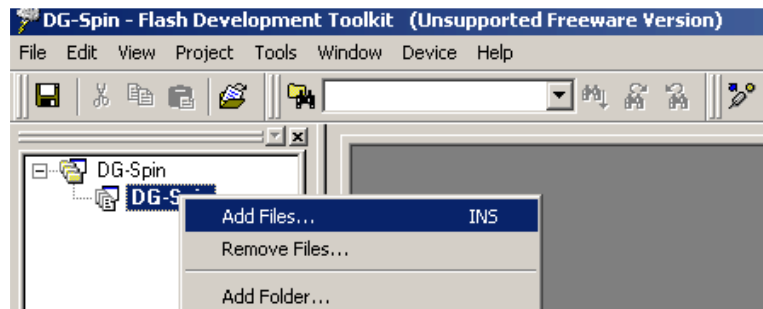


Figure 40. FDT Workspace

- Select the file with the *DG Spin* firmware from the suitable folder and click *Add*.

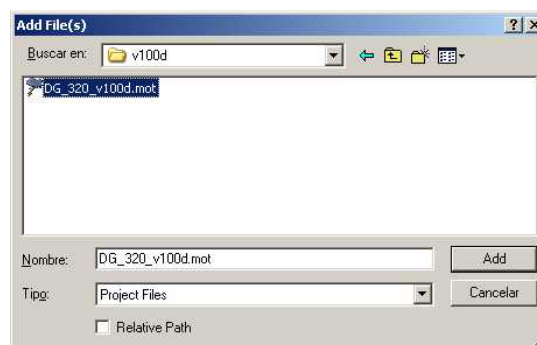


Figure 41. Add file window of the FDT

- Check that the centrifuge is switched on and all the connections properly attached. Press the *right mouse button* on the file name and select the *Download File* option at the following screen:

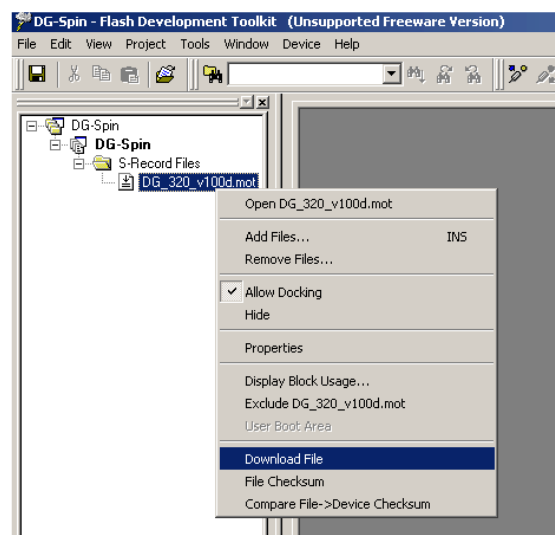


Figure 42. Download file option in FDT Workspace

- The FLASH programming software will program the *DG Spin* memory. The programming progress will be shown on the lower window. Should any error occur, the software will issue a message error. In this case, check that the board is connected, turn off the centrifuge and try the operation again.

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- The recording has been completed. Switch off the *DG Spin*, **remove the connection between the pads of the jumper JP3 from the PCA**, and switch on the equipment again to check that the new version is working properly. You can check the firmware version with *DG Terminal*.


WARNING!

It is recommended to turn off the instrument before performing the replacement procedures.


WARNING!

To completely disconnect the apparatus from mains remove the mains plug from the socket outlet or the appliance coupler.

6.2.3 Replacement of the PCA Distribution or PCA DC Motor Control

1. Open the centrifuge.
2. Remove the PCAs from the base.
3. Disconnect the PCAs and replace the damaged one.
4. Place the PCAs in the base.
5. Be sure that cables are properly connected.
6. Close the centrifuge.
7. Perform an electric continuity test
8. If the PCA replaced is the PCA DC Motor Control, install the latest firmware version.
9. Execute the OQ procedure to assure that the centrifuge *DG Spin* works properly.

6.2.4 Replacement of the motor

1. Open the centrifuge.
2. Unscrew the shock absorbers to remove the motor from the case.
3. Remove the shock absorbers, the motor cover and the PCA Unbalance Detector from the damaged motor and place them in the new one.
4. Screw the shock absorbers to the case.
5. Be sure that cables are properly connected.
6. Close the centrifuge.
7. Perform an electric continuity test
8. Execute the OQ procedure to assure that the centrifuge *DG Spin* works properly.

7 ADDITIONAL INFORMATION

7.1 Log messages

A complete overview of all possible log events is shown in the table below:

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NORMAL OPERATION	
Log message	Explanation
Abort centrifugation	Centrifugation aborted by user.
Boot and reset	An external boot command has reset the device.
Device ON	Device switched on.
Device OFF	Device switched off.
End centrifugation	Centrifugation has ended.
Lid closed	Lid closed.
Lid opened	Lid opened.
Firmware version: version and date	Actual firmware version installed.
Firmware update	A new firmware version has been installed.
Start centrifugation	Start of centrifugation.

Table 3. List of log messages during normal operation of the DG Spin

CONFIGURATION MODIFICATION	
Log message	Explanation
Configuration change: Buzzer Volume	Buzzer volume changed.
Configuration change: Time	Time and date changed.
Configuration change: Units	Units changed between rpm and g.

Table 4. List of configuration modification log messages of the DG Spin

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WARNINGS	
Log message	Explanation
CURRENT LIMIT	The motor consumption was higher than the programmed.
DOOR OPEN WARNING	The lid has been opened while the motor was moving.
LOCK FAILURE WARNING	Failed to open the lid 2 consecutive times.
MOVEMENT WARNING	The difference between the actual motor position and calculated one is beyond limits (current limit).
NO ROTOR PRESENCE	A centrifugation without rotor has been performed and it has been aborted.
REGULATION WARNING	The regulation speed of the motor is beyond its margins (± 50 rpm).
RTC OUT OF BATTERY	The Real Time Clock battery is low and it has lost the date and time stored.
VIBRATION SENSOR WARNING	Failed to communicate with the vibration sensor.
VIBRATION WARNING	Excessive level of vibration detected.

Table 5. List of warning log messages of the DG Spin

SYSTEM ERRORS	
Log message	Explanation
EEPROM ERROR (E3)	I2C communication fails with the EEPROM memory.
FPGA_ERROR (E9)	Failed to program the FPGA.
LCD ERROR (E2)	I2C communication fails with the LCD.
RTC ERROR (E1)	I2C communication fails with the Real Time Clock.

Table 6. List of system errors log messages of the DG Spin

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8 DRAWINGS

This chapter describes the available options when viewing the digital drawings representing all the modules of DG Spin instrument, including 3D views and animated assembly procedures that are performed in factory for each of them. As detailed below, a free software is required to view the drawings and assembly procedures, which are contained in '.pvz' files as an attachment to the current Service Manual document.

8.1 Requirements

- 3D drawings ('.pvz' files)

The file name format of the 3D drawings is *MxxxxxxVn_myy.pvz*, where *xxxxxx* is the module identification number, *n* is a number indicating the drawing version, and *yy* is the modification number for the corresponding module. 3D drawings are included in the *Drawings* folder as part of the current Service Manual.

- Drawings table

This table can be used to find the name of the drawing corresponding to each module.

- Spare parts list

The raw material codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.

- 3D viewer

Creo View Express is the software used to view the available 3D drawings and the corresponding assembly procedures. It can be downloaded following the instructions provided in section 8.3.1 *Installation* and configured as detailed in section 8.3.2 *Configuration*.

8.2 Drawings reference table

There are the following 3D drawings (pvz format):

Module	Description	Number
GEN	General assembly	M6007523V0_m04.pvz

Table 7. List of 3D drawings

Additionally, there are the following mechanical drawings (2D, pdf format):

Module	Description	Number
PACK	Packaging	S1248V0_m01.pdf
	De-assembly	S0799V0_m00.pdf
LID	Lid assembly	S0722V1_m05.pdf

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CABLING	Internal cabling	S0803V0_m06.pdf
	AC power cable	p1797v0_m04.pdf
	Ground cable	p1798v0_m02.pdf

Table 8. List of 2D drawings

8.3 Creo View Express

8.3.1 Installation

- **Within Grifols company**

Install **Creo View Express** from Grifols Software Center.

- **External distributors**

Look for Creo View Express in <https://www.ptc.com> (free viewer) and follow the downloading and installation instructions.



NOTE:

Use Creo View Express software version 8.0 (or higher).

8.3.2 Configuration

Once a drawing (‘.pvz’ file) is opened using Creo View Express, the main structure of that module can be displayed by selecting the *Structure* tab on the left panel of the screen, then double-clicking on the main structure code contained in the *Ilustración* field.



NOTE:

Some drawings may contain more than one main structure, so they should be opened individually.

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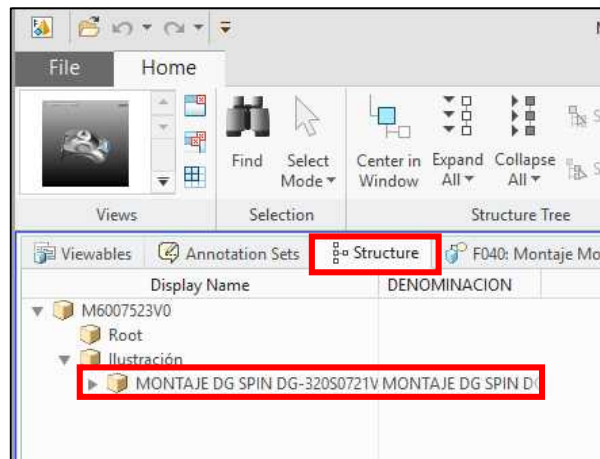


Figure 43. Main structure code

A new tab named *Model* will be created, and the 3D drawing of the selected module will be shown on screen. At this point, several drawing techniques can be used to represent the model by selecting the *Render Mode* button in Creo View Express toolbar (the shaded view is selected by default).

When selecting any of the parts composing the drawing, the corresponding component will be highlighted in the navigation tree on the left side of the screen. In order to see the raw material description and its code (factory reference) instead of the default values, the following configuration should be applied:



NOTE:

This configuration is required if raw material codes need to be searched within the module

1. Go to *File/Options*.
2. Go to *General/Navigation* window.
3. Enable the *Use alternate path names* checkbox and click on *Configure*. A new window *Configure Alternate Part Names* opens.
4. Click on *Insert Attribute....*
5. Select 'CODIGO' and click on *Insert value*.
6. Select 'DENOMINACION' and click on *Insert value*.
7. Consider applying a whitespace between both values in the *Configure Alternate Part Names* window.

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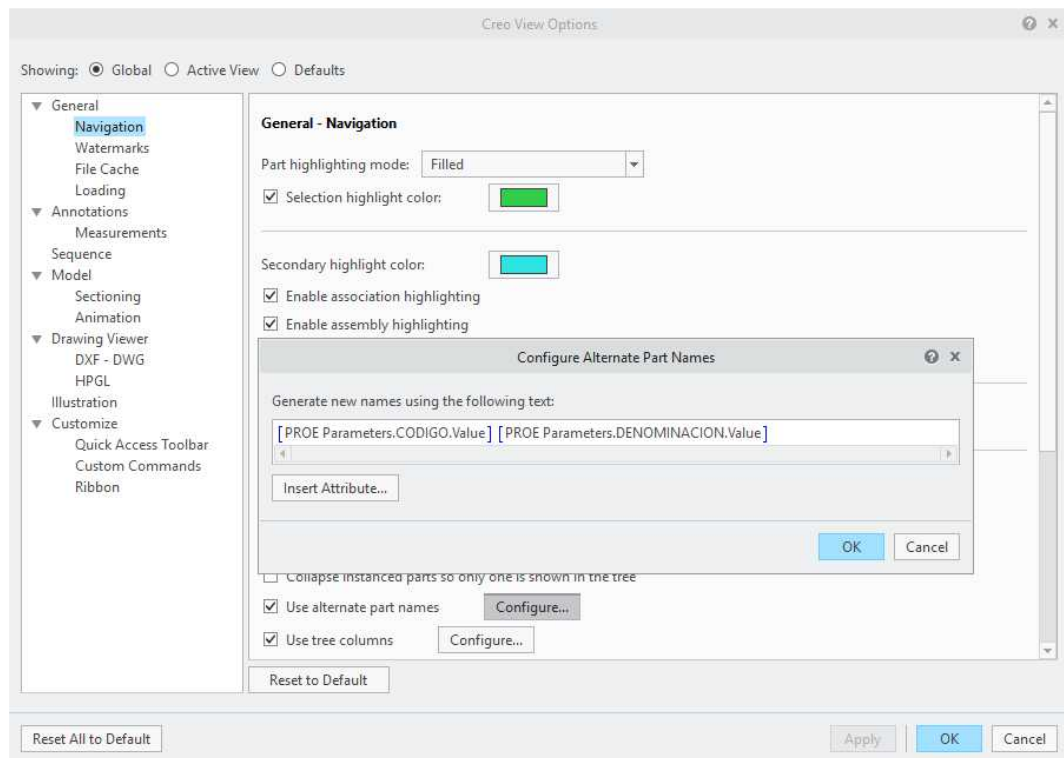


Figure 44. Creo View Options

8. Click on *Close*, then *OK*.
9. Click on *Apply*, then *OK*.
10. Now the raw material description and its factory reference can be seen in the navigation tree for each part composing the drawing. This reference can also be found when searching for *CODIGO* field.

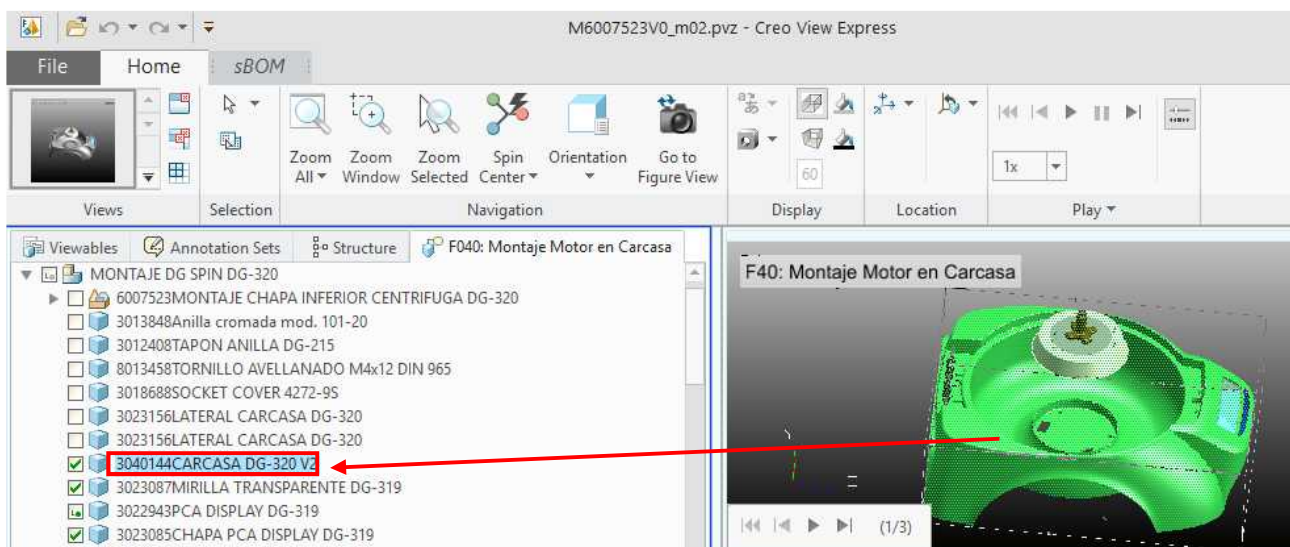


Figure 45. Highlighted part description after selecting a drawing part

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8.3.3 Models structure and navigation

8.3.3.1 Tree structure

1. Click on *Structure* tab and double click on the main structure code contained in the *Ilustración* field.

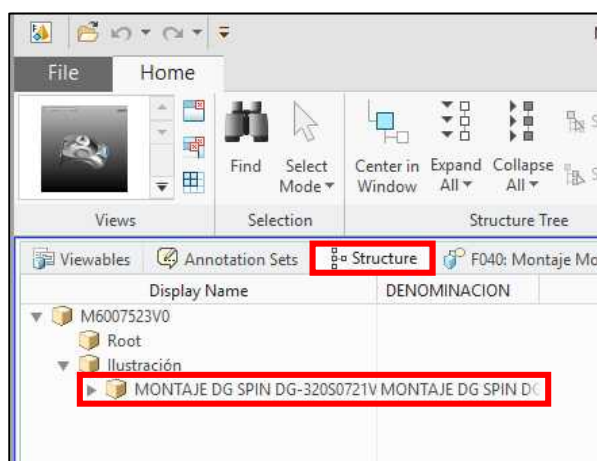


Figure 46. Main structure code

2. A new tab named *Model* will be created, and the main model of the selected module will be shown on screen. The tree structure (starting on the main structure code) can be expanded or collapsed by clicking on the triangles next to each field.
3. When double clicking in any field of the tree, the corresponding component in the drawing is zoomed in and changes its color to blue (subassembly containing raw materials) or green (raw materials).

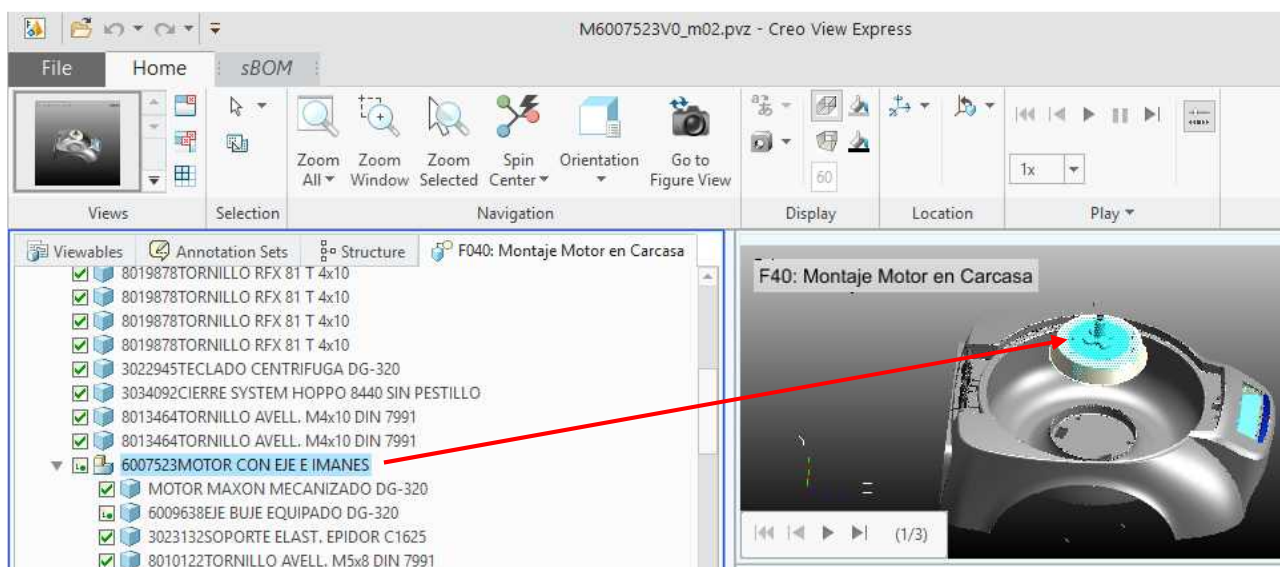


Figure 47. Tree structure component selection

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4. Parts can be hidden or isolated by right-clicking on them and selecting the desired action, and then reestablished by right-clicking anywhere within the drawing area and selecting *Unhide All*. They can also be hidden or unhidden by enabling/disabling the check square near each component in the tree structure.

8.3.3.2 Models navigation

Once the drawing is displayed on screen, it can be rotated, dragged, zoomed in or out, etc. by using the below listed functionalities for Creo View Express:

- Zoom in/out (pointer area taken as reference point):
 - Mouse: rotate the mouse wheel.
 - Pad: put two fingers on pad and move them up or down.
- Rotate model:
 - Mouse: Press and hold right button and drag mouse.
 - Pad:
 - Select *Free Rotation* button in Creo View Express toolbar.
 - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
 - Hold left click on the sphere while dragging on the pad.
- Drag model:
 - Mouse: Press the wheel and drag mouse.
 - Pad:
 - Select *Transform* button in Creo View Express toolbar.
 - Select in the *Ilustración* field in the tree structure, or just select a part of the drawing.
 - Hold left click on a sphere's center grey circle while dragging on the pad.

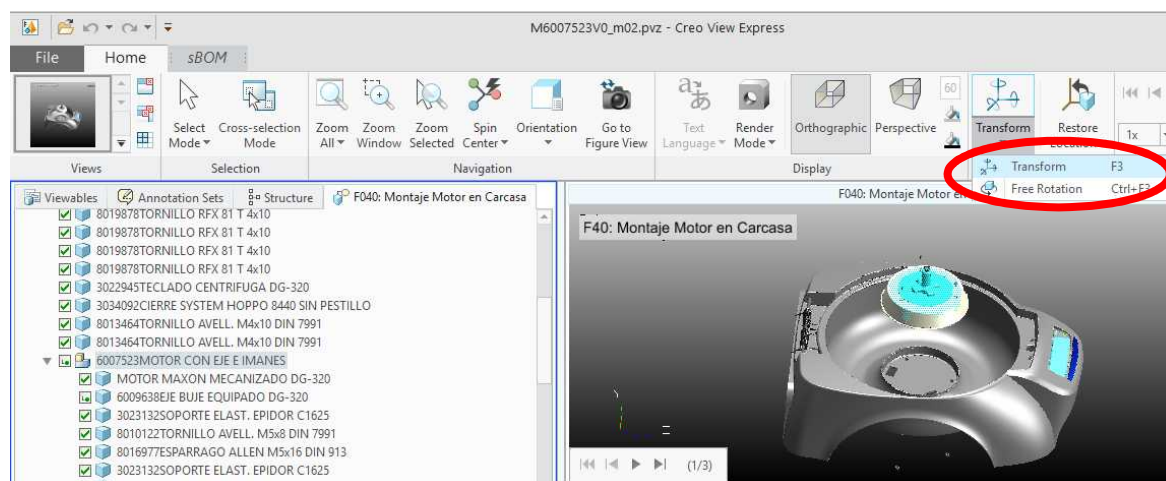


Figure 48. Transform options in Creo View Express toolbar

- Hide/Unhide part:
 - Select the part clicking by right-clicking on it and select *Hide* or *Unhide All*.
 - Select the part clicking on it and enable or disable it in the tree structure.
- Center model on screen: double click on the background

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- Restore location: click on *Restore location* button in Creo View Express toolbar. With this, any part that has been moved/rotated from its default position within the drawing area will restore its original position.

8.3.3.3 Animated assembly procedures

The *Viewables* tab contains a list of animated assembly procedures detailing the assembly steps as done in factory for each of the modules composing DG Spin instrument. Each component being assembled in an animation is identified with its raw material code.

- When opening a '.pvz' file for a specific module, the *Viewables* tab is selected by default, showing the corresponding animated assembly procedures (*Figures* area).

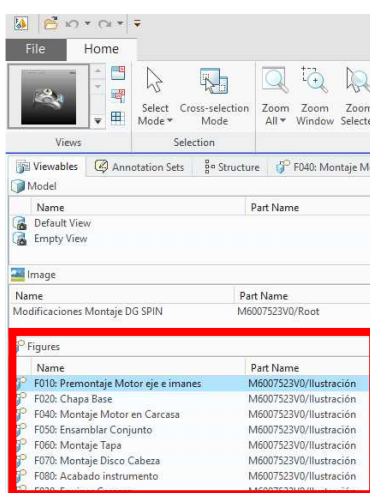


Figure 49. Figures area

- Double click on the desired assembly and a new tab will be created, showing the tree structure with the parts involved along with the corresponding animated assembly.
- The animation can be played and paused for each of the steps composing it by using the *Playback* control buttons at the bottom left corner of the animation screen, or those at the *Play* section in Creo View Express toolbar.

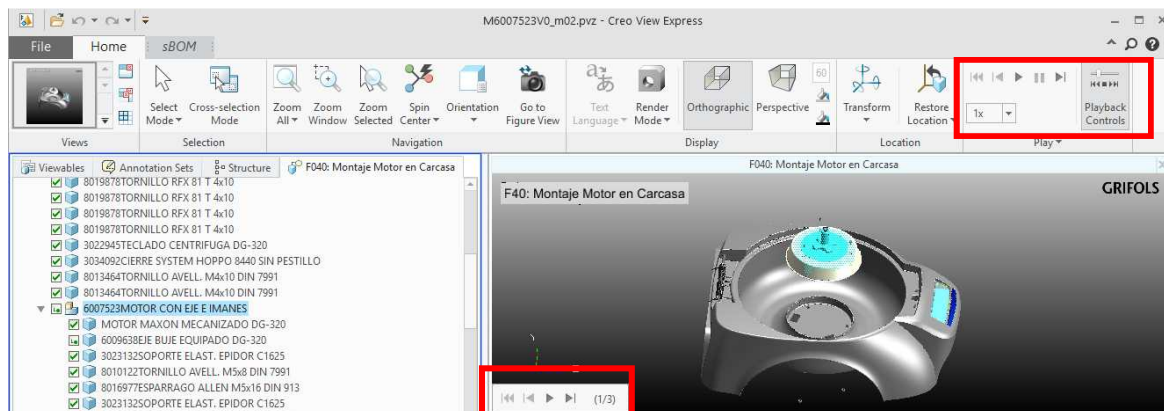


Figure 50. Playback control buttons for an animated assembly procedure

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- Parts can be hidden/unhidden during the animation by enabling/disabling the square near each component in the tree structure or selecting the corresponding option after right-clicking on it.
- The background colors can also be modified as desired by editing the *Background* and *Gradient* options of the *Display* section in Creo View Express toolbar.

8.3.3.4 Spare parts searching

The raw codes composing a factory module can be found in the 3D drawings. From there, they can be matched to the codes composing an existing spare part in order to know the corresponding spare part reference.



NOTE:

Before searching for a specific raw material code, make sure that the configuration detailed in section 8.3.2 *Configuration* has been applied

8.3.3.5 Finding a drawing part for a spare parts list's reference code

- Copy the desired drawing reference code from the Spare Parts List 'Drawing reference' column.
- Search in the drawings table the drawing of the module where the part is assembled.
- Click on *Find* button in Creo View Express toolbar.
- Paste the previously copied code in the new window and click on *Find next*.
- The raw material corresponding to the selected drawing reference will be highlighted both in the tree structure and in the model drawing.

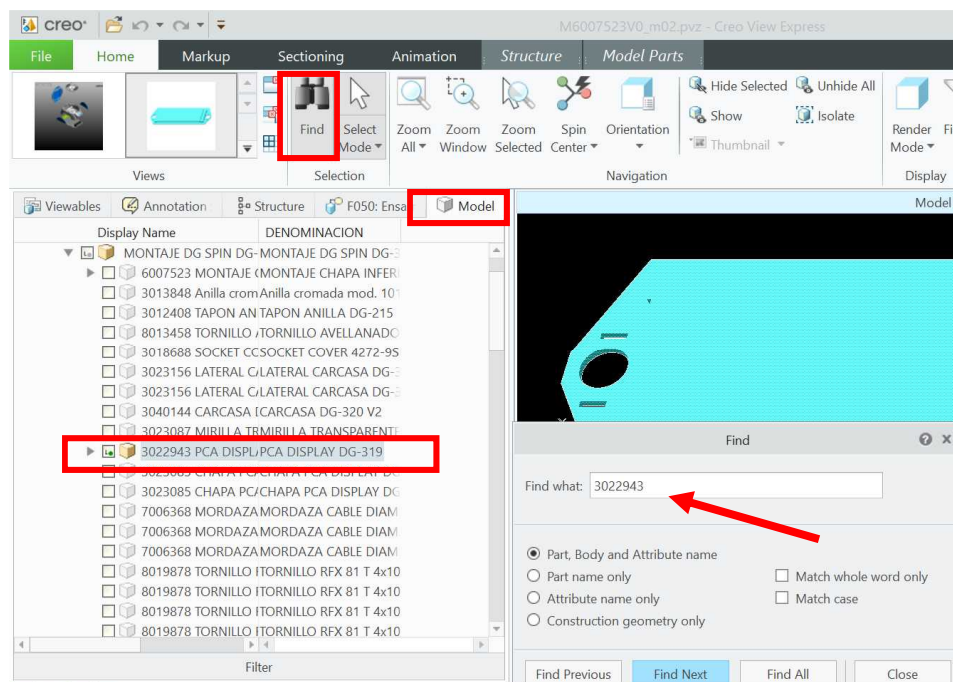


Figure 51. Finding a drawing part from a known code

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8.3.3.6 Finding the spare part list's reference code for a drawing part

1. Select the part of the drawing which raw material code needs to be known.
2. The part will be selected in the tree structure. If the view configuration detailed in section 8.3.2 *Configuration* 8.3.2 has been applied, the raw material code (CODIGO field) will be listed as part of the material description. The raw material code (CODIGO field) can also be seen by enabling the lower data panel option on the lower left side of the screen.

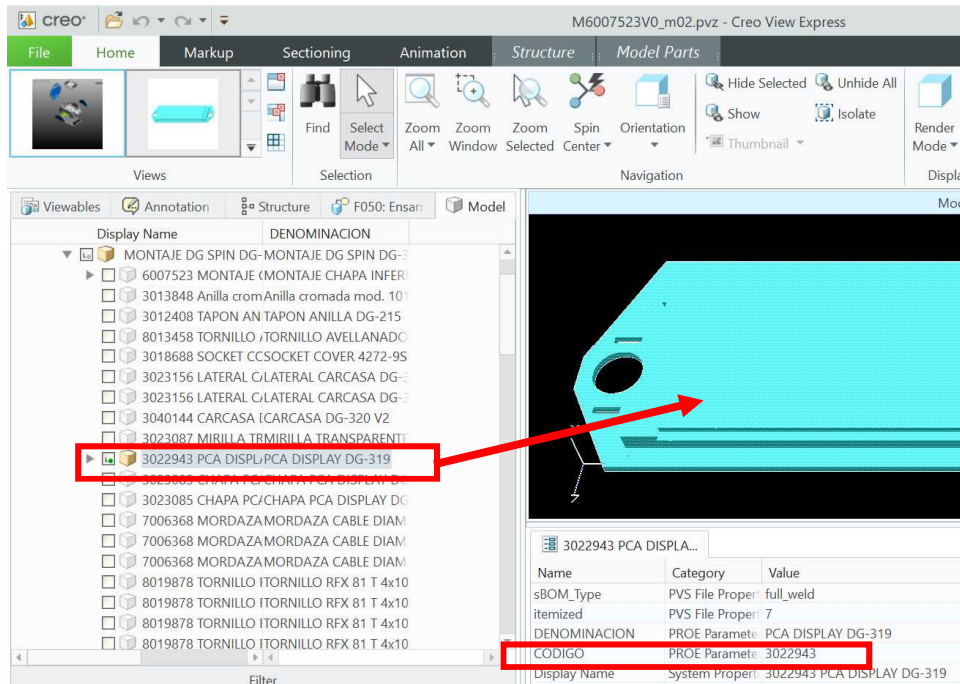


Figure 52. Find a code by selecting a drawing part

3. Search for the obtained raw material code in DG Spin Spare Parts List. The code will be listed in the *Drawing Reference* column for one or more spare part codes, so the desired reference can be obtained. If the raw material code cannot be found in the Spare Parts list:
 - a) If the raw material code is part of a set of codes included in a 'parent' code on top of it in the tree structure, try to look for the parent code in the Spare Parts list. The parent code of a specific part can be automatically selected by selecting it and then clicking on the *Select Parent* button than can be found in the *Structure* tab in Creo View Express toolbar.
 - b) If the raw material code is not part of a set of codes included in a 'parent' code, or if the parent code cannot be found in the Spare Parts list, it means that no part number reference has been assigned to the desired raw material code.